

Date: Sept 30, 2025

Math Trimester 1

MorningWork

Name:

Class of:

Teacher:

Homerroom:

8) When solving $x^2 + 6x = -8$ for x , a student wrote $x^2 + 6x + 8 = 0$ as their first step. Which property justifies this step?

- (1) associative property
- (2) commutative property
- (3) zero property of addition
- (4) addition property of equality

Solution: We add 8 to both sides of the equation to obtain $x^2 + 6x + 8 = 0$. This is the addition property of equality. Choice (4)

9) The inputs and outputs of four different functions are listed below (as ordered pairs).

$$f(x) : (-2, 6), (-1, 1), (0, -2), (1, -3), (2, -2), (3, 1)$$

$$g(x) : (-4, 3), (-3, 2), (-2, 1), (-1, 0), (0, 1), (1, 2)$$

$$h(x) : (-2, -1), (-2, -2), (0, -4), (1, -8), (2, -16), (3, -32)$$

$$j(x) : (-3, -11), (-2, -7), (-1, -3), (0, 1), (1, 5), (2, 9)$$

Which list represents a linear function?

- (1) $f(x)$
- (2) $g(x)$
- (3) $h(x)$
- (4) $j(x)$

Solution: Check first differences in y for consecutive x -values. For $j(x)$: $-11, -7, -3, 1, 5, 9$ increases by $+4$ each step, so slope is constant $\Rightarrow$ linear. Others do not have a constant rate of change (and h even assigns two outputs to $x = -2$). Choice (4)

16) Which equation is always correct?

- (1) $a^3 \cdot a^x = a^{3x}$
- (2) $(a^4)^x = a^{4+x}$
- (3) $(ab)^x = a^x b^x$
- (4) $a^x \cdot b^y = ab^{x+y}$

Solution: Exponent rules give $a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, and $(ab)^x = a^x b^x$. Only statement $(ab)^x = a^x b^x$ matches a correct rule as written. Choice (3)

18) Three functions are given below:

$$f(x) = -|x + 2| + 7, \quad g(x) = (x - 3)^2 - 4, \quad h(x) \text{ has } h(0) = -3.$$

(Additional values for h : $(-4, 5), (-3, 0), (-2, -3), (-1, -4), (0, -3), (1, 0), (2, 5)$.)

Which functions have the same y -intercept?

- (1) $f(x)$ and $g(x)$
- (2) $g(x)$ and $h(x)$
- (3) $f(x)$ and $h(x)$
- (4) The functions all have different y -intercepts

Solution: Evaluate at $x = 0$. $f(0) = -|2| + 7 = 5$. $g(0) = (0 - 3)^2 - 4 = 9 - 4 = 5$. $h(0) = -3$. Thus f and g share the same y -intercept 5. Choice (1)

24) In 2009, Usain Bolt set the world record in the 100-meter dash with a time of 9.58 seconds. His approximate speed, in kilometers per hour, can be found using which conversion?

- (1) $\frac{9.58 \text{ sec}}{100 \text{ m}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot \frac{1 \text{ min}}{60 \text{ sec}} \cdot \frac{1 \text{ hr}}{60 \text{ min}}$
- (2) $\frac{100 \text{ m}}{9.58 \text{ sec}} \cdot \frac{60 \text{ sec}}{1 \text{ min}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot \frac{60 \text{ min}}{1 \text{ hr}}$
- (3) $\frac{100 \text{ m}}{9.58 \text{ sec}} \cdot \frac{1 \text{ km}}{1000 \text{ m}} \cdot \frac{1 \text{ min}}{60 \text{ sec}} \cdot \frac{1 \text{ hr}}{60 \text{ min}}$
- (4) $\frac{100 \text{ m}}{9.58 \text{ sec}} \cdot \frac{60 \text{ sec}}{1 \text{ min}} \cdot \frac{1 \text{ km}}{1000 \text{ m}} \cdot \frac{60 \text{ min}}{1 \text{ hr}}$

Solution: Start with $\frac{100 \text{ m}}{9.58 \text{ s}}$ and convert meters $\rightarrow$ kilometers using $\frac{1 \text{ km}}{1000 \text{ m}}$, then seconds $\rightarrow$ hours using $\frac{60 \text{ s}}{1 \text{ min}} \cdot \frac{60 \text{ min}}{1 \text{ hr}}$. This matches option (4).